

Liam Meagher

+1 (480) 980-7007 | liammeagher03@gmail.com | linkedin.com/in/liammeagher03 | Los Angeles, CA | US Citizen

EDUCATION

University of Southern California (USC)

M.S. in Mechanical Engineering - *Expected May 2028*

University of California, Los Angeles (UCLA)

B.S. in Computer Science & Engineering

RELEVANT EXPERIENCE

Drivetrain, Electronics, and Armor Lead Engineer *Flagship Combat Robotics*

2024-2025

- Designed and manufactured a **30 lb** autonomous vertical spinner according to a provided schematic's constraints
- Leveraged Finite Element Analysis to iterate a weapon component into a subassembly supporting **3000 N** of force
- Fabricated and assembled several rapid prototypes of various assemblies using laser cutters and additive 3D-printing
- Instructed **26 team members** on researching, designing, and machining **2 competition-ready combat robots**
- Manufactured critical steel and aluminum components using 3D printing, manual mill, lathe, and CNC machining

Autonomous Systems Software Developer *Bruin Underwater Robotics*

2023-2024

- Designed and implemented a feedback control system and codebase enabling autonomous underwater navigation
- Generated synthetic training datasets in Unity with ground-truth physics for the vehicle's computer vision model
- Integrated software with mechanical and electrical systems to ensure reliable and robust system-level functionality

Full-Stack Web Developer *BruinSwap: UCLA Marketplace*

2022-2023

- Designed and developed a peer-to-peer online marketplace enabling UCLA students to buy, sell, and trade items
- Implemented and maintained SQL databases for user authentication, account management, and digital asset storage
- Oversaw Git workflows for JavaScript codebase, resolving merge conflicts and maintaining the team's code integrity
- Enabled **150+ users** to trade securely on UCLA's first exclusive student-oriented marketplace without user fees
- Achieved **98%** uptime with Node.js landing page and secure peer-to-peer messaging board across student accounts

Activities and Outdoor Programming Director *Scouting America*

2021-2022

- Operated a weekly outdoor activity program supporting over **4,000 participants** with a budget of over **\$390,000**
- Directed, evaluated, and appraised **30 staff members** and provided constructive feedback to develop their skillsets
- Maintained a varied inventory of program equipment to support and grow staff morale of **170+ full-time staff**
- Created a thorough end-of-year report for future directors to track and monitor responsibilities between seasons
- Recruited, interviewed, hired, and trained relevant staff members for subsequent 2022 and 2023 summers

PROJECTS

Lightweight Combat Robotics *ASME at UCLA*

2022-2025

- Designed and manufactured **3 lb** horizontal spinners to compete against universities and hobbyists in California
- Modeled in SolidWorks and manufactured chassis, weapon, and drivetrain via 3D printing, laser cutting, and soldering

C++ Game Developer *Adaptive Battleship AI*

2023-2024

- Designed and implemented a local Battleship executable featuring **4 scaling difficulty levels**, from beginner to expert
- Developed AI algorithms to analyze player patterns and reliably achieve win rates of **60-95%** across skill levels
- Introduced novel gameplay mechanics, including **5 special abilities**, to enhance replayability and player engagement
- Optimized performance and memory management to support large-scale simulation testing of AI strategies

Programming and Gameplay Mechanics Lead *Phemie's Fable*

2021-2022

- Developed and implemented classes, structures, and colliders for **4 weapon systems** and **16 unique animations**
- Designed a rotating weapon and stats system, creating dynamic gameplay diversity and increasing player engagement
- Collaborated in Unity on rapid prototyping, general debugging, and asset integration under tight time constraints

Project Lead *Eagle Scout Award Service Project*

2018-2019

- Designed and implemented a drainage system with civil engineers to redirect floodwater to nearby storm drains
- Fundraised **over \$2000** and led **28 volunteers** in moving **4,000+ lbs** of gravel to reduce local flooding and erosion

SKILLS

Languages: C, C++, C#, Go, Haskell, Java, JavaScript, Linux, Python, R, Swift, Verilog

Tools: Docker, Matplotlib, Node.js, NumPy, OpenCV, Pandas, PyTorch, React, Scikit-Learn, Solidworks, Unity, Unreal Engine

Machining Skills: 3D Printing, Automatic Tapping, CNC Machine, Drill Press, Laser Cutting, Lathe, Mill, Resin Printing, Soldering